Steven Vu

stevenquvu@gmail.com | linkedin.com/in/stevvu | github.com/STEVVVU

Education

Georgia Institute of Technology 2028
M.S. in Computer Science

University of Houston 2026
B.S. in Computer Science | *GPA: 3.8/4.0*

Experience

Amazon June 2026 – Sept. 2026
Software Development Engineer Intern *Palo Alto, CA*

- Built self-service YAML workflows on Amazon EMR and Apache Spark, orchestrated with AWS Step Functions.
- Developed a React interface for scientists to launch, monitor, and approve research experiments.
- Cut research iteration time 50% using an Amazon Bedrock LLM agent for interactive data exploration.
- Automated Spark data validation and AWS CloudWatch alerts, replacing manual notebook quality checks.

Amazon June 2025 – Sept. 2025
Software Development Engineer Intern *Seattle, WA*

- Built Java/Redis frequency caps for Amazon Search Autocomplete, serving 10M+ daily customers.
- Processed 10M+ hourly impression events from Amazon S3 using Apache Spark on Amazon EMR.
- Orchestrated hourly pipelines with AWS Step Functions and EventBridge; provisioned via CloudFormation.
- Kept Java rule evaluation under 25 ms p99 with Redis TTLs and fail-open handling; 99.9% uptime.

Roblox Feb. 2022 – Present
Game Developer *Remote*

- Built multiplayer and monetization systems for games with 700K concurrent players and $8M+ revenue.
- Reduced multiplayer combat latency with compact network updates and client-side prediction.
- Prevented conflicting player-state writes using session locks and distributed in-memory caching.
- Blocked multiplayer exploits with server-side hit validation and request rate limiting.

Projects

Document Data Warehouse | *Gemini, BigQuery, Cloud Storage, SQL*

- Built an AI-assisted invoice warehouse with Gemini extraction, Cloud Storage staging, and BigQuery analytics.
- Designed 12 SQL data models with source hashes to trace supplier and spending reports back to invoices.
- Used versioned updates and deletion markers to preserve exact currency totals across 20 local test runs.

Parallel Task Scheduler | *C++, Multithreading, Work Stealing*

- Extended a C++ task scheduler with worker-affinity hints and work stealing for parallel workloads.
- Used atomic task claims across per-worker mailboxes and stealable queues to prevent duplicate execution.
- Ran a 131K-cell, 256-step stencil in 243.9 ms on 4 workers vs. 696.5 ms on 1 with affinity hints.

Collaborative Python Trace Notebook | *React, TypeScript, WebSockets, SQLite*

- Built a React/TypeScript Python notebook for live code collaboration, execution tracing, and offline editing.
- Synced 32 test peers at 50 edits/s using CRDTs and WebSockets; median p99 convergence was 15.52 ms.
- Persisted state for restart recovery; completed 141 protocol cases with 29,150 edits and no failures.

Technical Skills

Languages:	Python, Java, C++, TypeScript, JavaScript, Go, SQL, Rust
Web & APIs:	React, Node.js, FastAPI, HTTP APIs, WebSockets, gRPC
Cloud & Infra:	AWS (S3, EMR, Step Functions, ElastiCache), GCP (BigQuery, Cloud Run), Docker, Kubernetes, Terraform
AI & Data:	Apache Spark, Redis, Amazon Bedrock, Gemini, Firestore, PyTorch, PostgreSQL